

1. Consider the list 12, 3, 13, 6, 3, 1, 12.

- a. Use a bubble sort to sort this list. Show the lists at the end of each 'pass' of the algorithm and count the total number of comparisons you did in each pass.

12	3	13	6	3	1	12	Comparisons

- b. Use a shuttle sort to sort this list. Show the lists at the end of each 'pass' of the algorithm and count the total number of comparisons you did in each pass.

12	3	13	6	3	1	12	Comparisons

2. Consider the list 4, 9, 1, 3, 2, 7, 5, 6, 8.

- a. Show the result of the first three passes when the bubble sort is used to sort the list

4	9	1	3	2	7	5	6	8

- b. Explain why there will be exactly one additional pass before the algorithm terminates, and state the number of comparisons that will be needed during this final pass.

3. Five integers are to be sorted into ascending order.
A bubble sort is to be used on the list 6, 4, x , 2, 11. After the first pass, the list becomes 4, x , 2, 6, 11.
After the second pass, the list becomes x , 2, 4, 6, 11.
A shuttle sort is now used on the list 11, x , 2, 4, 6. After the first pass, the list of numbers becomes x , 11, 2, 4, 6. After the second pass, the list becomes 2, x , 11, 4, 6.
Find the value of x .
4. Find a list of ten numbers with the property that a bubble sort will terminate after exactly 24 comparisons.
5. A bubble sort is being used to sort a list of five numbers. After the first pass, the list is 3, 1, 6, 2, 9.
Find all the possibilities for the original list.
6. A shuttle sort is being used to sort a list of seven numbers. After the second pass, the list is 3, 5, 9, 2, 8, 1, 7. How many possibilities are there for the original list? Briefly describe them.
7. A permutation of the numbers 1 to 5 is being sorted using a shuttle sort, and exactly five comparisons are required. Find all the possibilities for the original list.
8. Use the bubble and shuttle sorts to sort each of the lists given below. In each case count the total number of comparisons that need to be made, and the total number of swaps that are made.

List	Bubble Sort		Shuttle Sort	
	Comparisons	Swaps	Comparisons	Swaps
2, 3, 4, 5, 6, 1				
6, 1, 2, 3, 4, 5				
4, 2, 3, 5, 6, 1				
1, 2, 1, 3, 2, 3				
2, 1, 4, 3, 6, 5				

9.
 - a. Explain why the number of swaps in a bubble or shuttle sort is never greater than the number of comparisons.
 - b. Determine the lists of length n which have the property that the number of swaps with the bubble sort is the same as the number of comparisons.
 - c. Determine the lists of length n which have the property that the number of swaps with the shuttle sort is the same as the number of comparisons.
10. For a particular permutation of the numbers 1 to n , the bubble sort requires $\frac{1}{2}n(n-1)$ comparisons to fully sort the list. What is the minimum possible number of swaps that could have taken place?
11. For a particular permutation of the numbers 1 to n , the shuttle sort requires $\frac{1}{2}n(n-1)$ comparisons to fully sort the list. What is the minimum possible number of swaps that could have taken place?
12. Prove that for any given list, the number of swaps required to sort this list using the bubble sort will always be equal to the number of swaps required using the shuttle sort.
13. [Challenge] Prove that for any given list, the number of comparisons required by the shuttle sort will never exceed the number of number of comparisons required by the bubble sort.